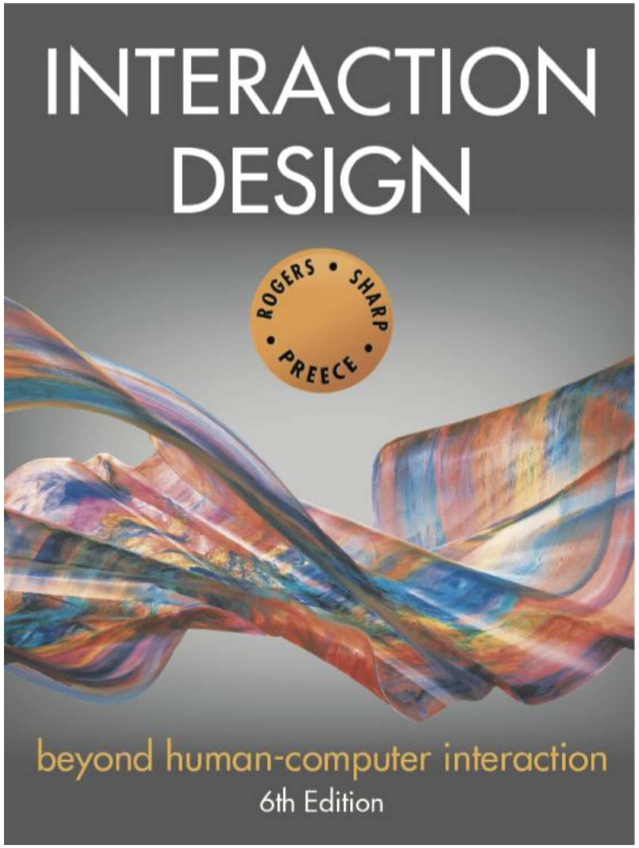


Chapter 7
INTERFACES

Overview

- Provide an overview of the diversity of interfaces
- Highlight the main design considerations for each of the interfaces
- Discuss what is meant by a natural user interface
- Consider which interface is best for a given application or activity

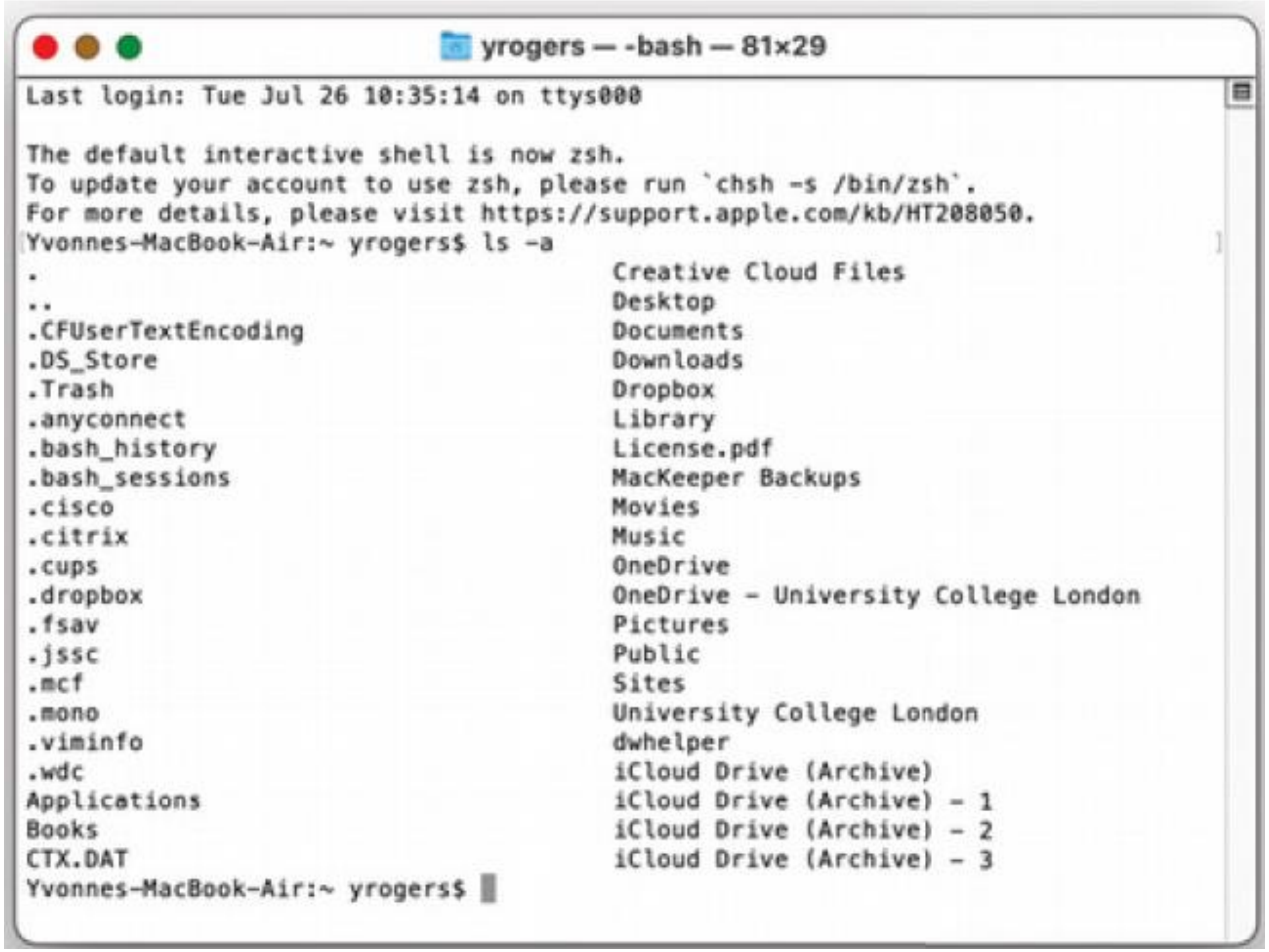
22 interface types covered

1. Command
2. Graphical
3. Multimedia
4. Virtual reality
5. Web
6. Mobile
7. Appliance
8. Voice
9. Pen
10. Touch
11. Touchless
12. Haptic
13. Multimodal
14. Shareable
15. Tangible
16. Augmented reality
17. Wearables
18. Robots and drones
19. Brain-computer
20. Smart
21. Shape-changing
22. Holographic

Command line interfaces

- Commands such as abbreviations (for instance, **ls**) typed in at the prompt to which the system responds (for example, by listing current files)
- Another way of issuing commands is by pressing certain combinations of keys (such as Ctrl+V).
- Some commands are fixed part of the keyboard such as delete, enter, escape, while some keys can be assigned to specific commands (e.g., F11 for print action)
- Efficient, precise, and fast
- Large overhead to learning set of commands

Command ls - lists what is on the laptop drive



```
yrogers — -bash — 81×29
Last login: Tue Jul 26 10:35:14 on ttys000

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
Yvonnes-MacBook-Air:~ yrogers$ ls -a
.                                       Creative Cloud Files
..                                      Desktop
.CFUserTextEncoding                     Documents
.DS_Store                               Downloads
.Trash                                  Dropbox
.anyconnect                             Library
.bash_history                           License.pdf
.bash_sessions                          MacKeeper Backups
.cisco                                  Movies
.citrix                                 Music
.cups                                   OneDrive
.dropbox                                OneDrive - University College London
.fsav                                   Pictures
.jssc                                   Public
.mcf                                    Sites
.mono                                   University College London
.viminfo                                dwhelper
.wdc                                    iCloud Drive (Archive)
Applications                            iCloud Drive (Archive) - 1
Books                                   iCloud Drive (Archive) - 2
CTX.DAT                                 iCloud Drive (Archive) - 3
Yvonnes-MacBook-Air:~ yrogers$
```

A Unix terminal display showing directories listed alphabetically for the command ls -a

Command line interfaces

- System administrators, programmers, and power users often find that it is much more efficient and quicker to use command languages such as Microsoft’s PowerShell.
- For example,
  - it is much easier to delete 10,000 files in one go by using one command rather than scrolling through that number of files and highlighting those that need to be deleted.

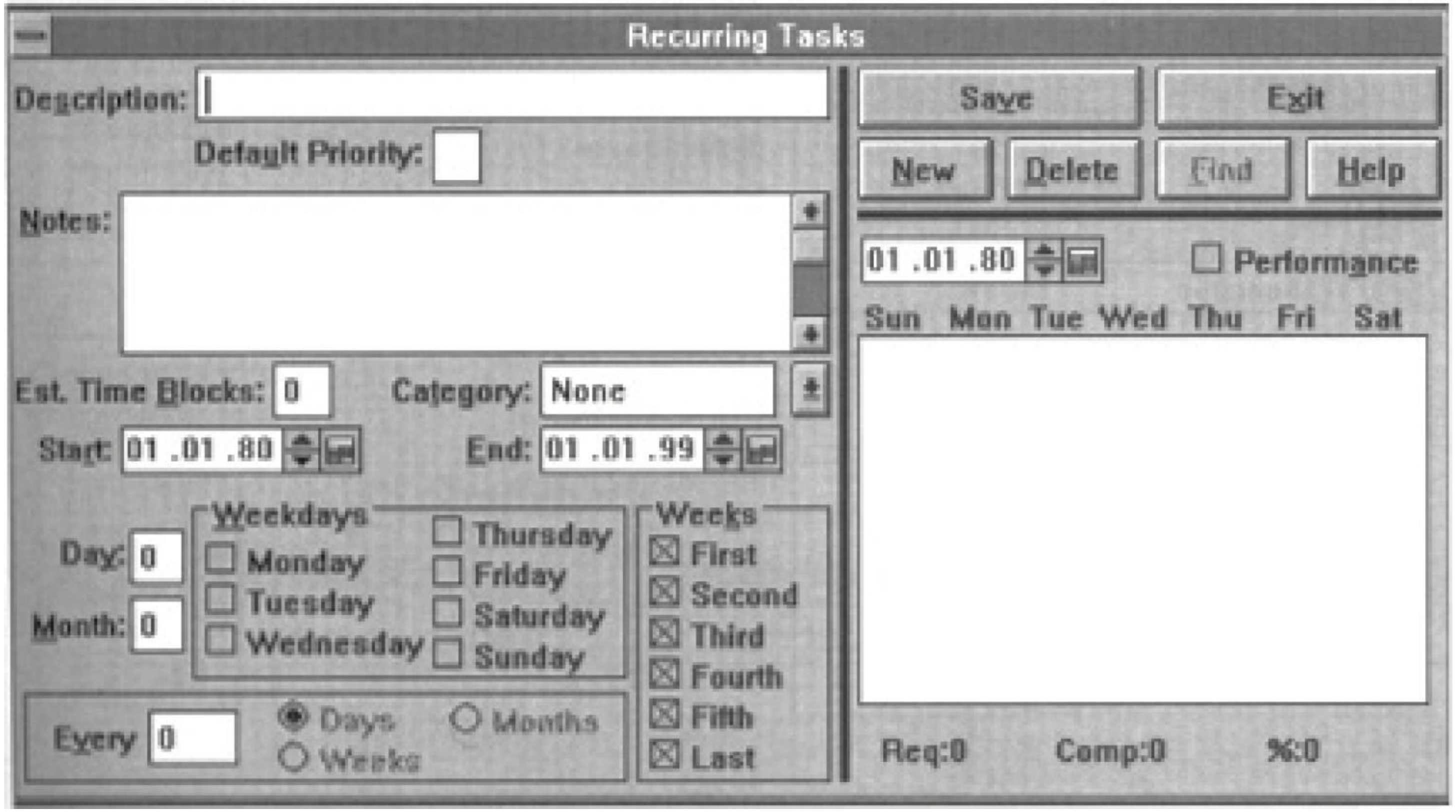
Design considerations

- Consistency is most important design principle (See Chapter 1)
  - The method used for labelling/naming the commands should be chosen to be as consistent as possible;
  - For example, always use first letter of command

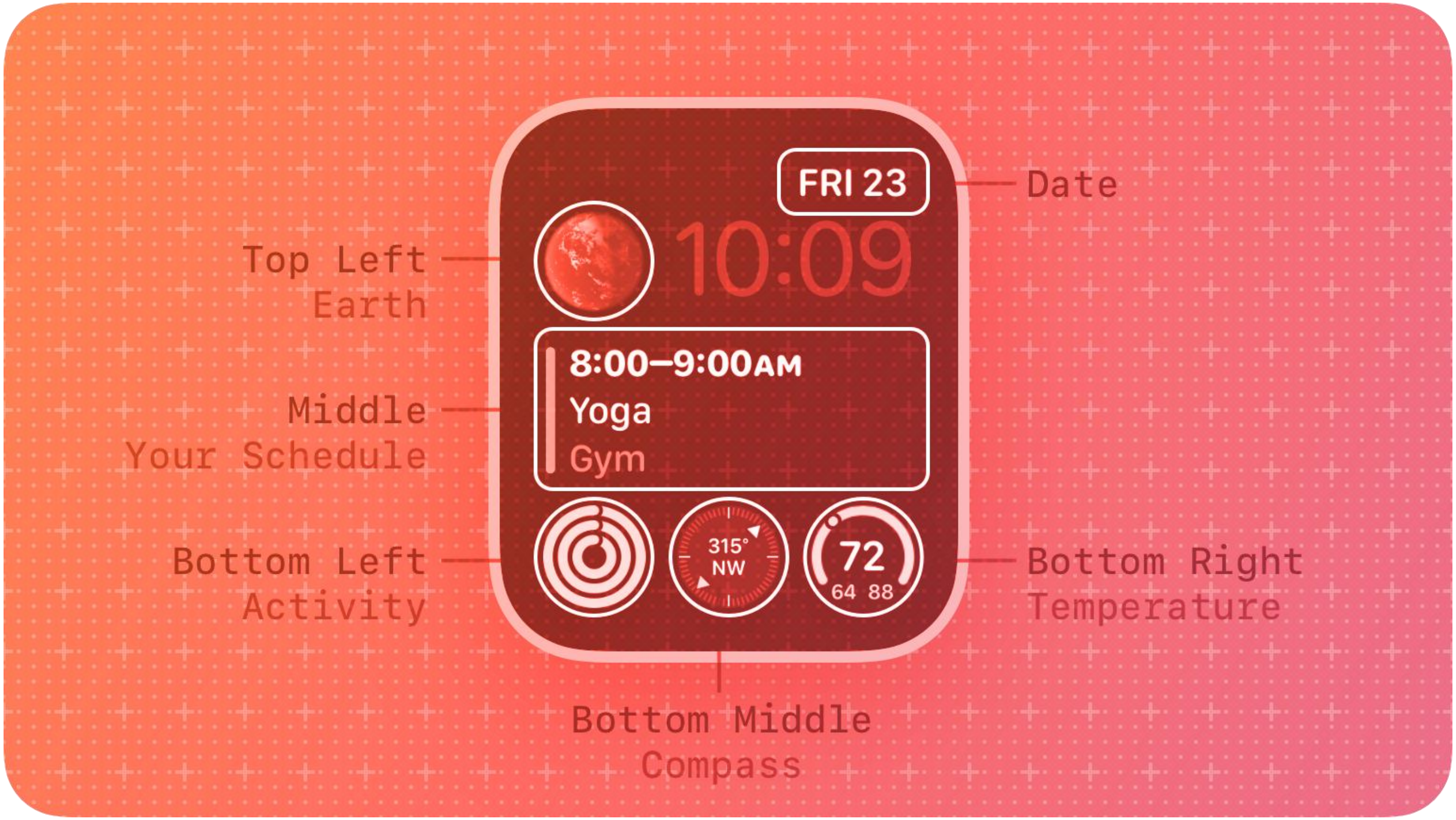
Graphical user interfaces (GUIs)

- Xerox Star first WIMP led to the birth of GUIs
- The original GUI was called a WIMP (windows, icons, menus, pointer) and consisted of the following:
  - Windows
    - Sections of the screen that can be scrolled, stretched, overlapped, opened, closed, and moved around the screen using the mouse
  - Icons
    - Pictograms that represent applications, objects, commands, and tools that were opened when clicked on
  - Menus
    - Lists of options that can be scrolled through and selected
  - Pointing device
    - A mouse controlling the cursor as a point of entry to the windows, menus, and icons on the screen

Example of first generation GUI



Smartwatch complication display



An Apple watchOS complication display that include the features “circulars” (three kinds shown in the bottom line) and “inlines” (shown in the middle and the upper right hand corner of the display)

Smartwatch complication display

- Complications are typically small widgets that can show various types of information
  - such as weather, fitness activity, upcoming calendar events, and more.
- A smartwatch complication display is the area on the watch face where these complications are shown

Windows

- The basic building block on the desktop and laptop is the window
  - in which apps, browsers, dialog boxes, interactive forms, feedback/error messages, etc., are presented.

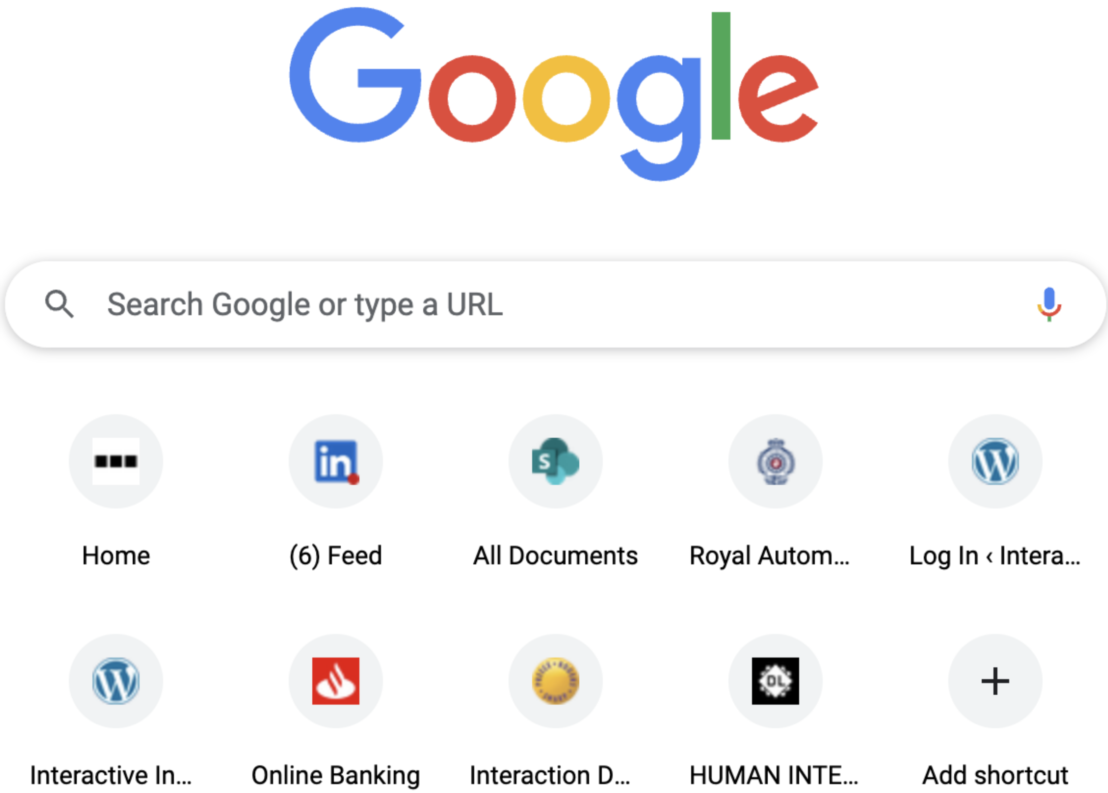
Window design

- Windows were invented to overcome the physical constraints of a computer display
- They enable more information to be viewed and tasks to be performed
  - Multiple windows can be opened at any one time,
    - for example, web browsers, word processing documents, photos, and slideshows,
    - User can switch between the windows when needing to look at or work on different documents, files, and apps.
  - Scroll bars within windows enable more information to be viewed
    - Enable upward, downward, and sideways movements through a document
    - How scrolling works on touch interfaces?

Window design

- Windows allow users to interact with multiple programs or applications simultaneously by displaying them side by side or overlapping each other on the screen.
- Multiple windows can make it difficult to find a particular window
  - Listing, tabbing, and thumbnails are techniques that can help

Window design: favicons of top sites visited below the Google search bar

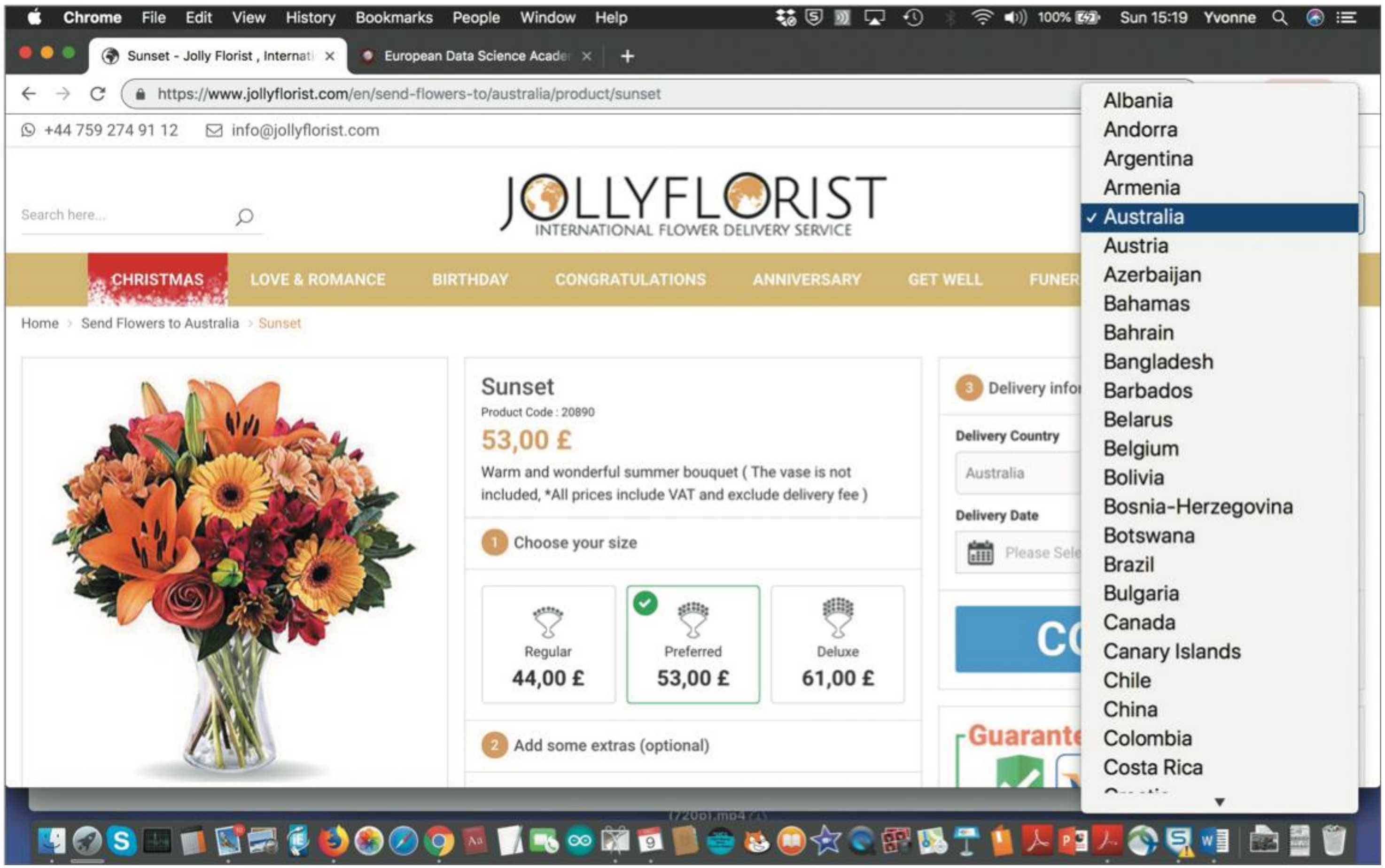


Web browsers, like Google and Firefox, also show a selection of favicons (i.e., favorite icons) of the top sites visited

The Joy of Filling in Forms Online

- One complaint about online registration forms is
  - the country of residence box that opens up as a never-ending dropdown menu, listing all of the countries in the world in alphabetical order.
  - Instead of typing in the country in which they reside, people are required to select the one they are from

Selecting a country from a scrolling window



A scrolling menu of country names.

What if you live in Venezuela?

The Joy of Filling in Forms Online

- A better design is to have a predictive text option ,
  - where someone needs only to type in the first one or two letters of their country to cause a narrowed-down list of choices to appear from which they can select within the interface.

Is this method better? If so why?

All Countries

A B C D E F G H I J K L M N O P Q R S T U V W XYZ

Algeria	Angola	Antigua
Argentina	Armenia	Aruba
Australia	Austria	Azerbaijan
Azores		

A section of the listing of countries in shaded blocks in alphabetical order, below a row of alphabetically ordered letters

Research and design considerations

- A key research focus is window management
  - Finding ways to enable users to move smoothly between different windows (and monitors)
    - How to switch attention between windows without getting distracted
  - Keyboard shortcuts and taskbars are often used for switching between windows.
  - Another technique is the use of tabs that appear at the top of the web browser that show the name and logo of the web pages that have been visited.
    - This mechanism enables people to rapidly scan and switch among the web pages they have visited.

Menu styles

Flat list: Good for showing small number of options at the same time when display is small

Drop down: Shows more options on same screen (for example, cascading)

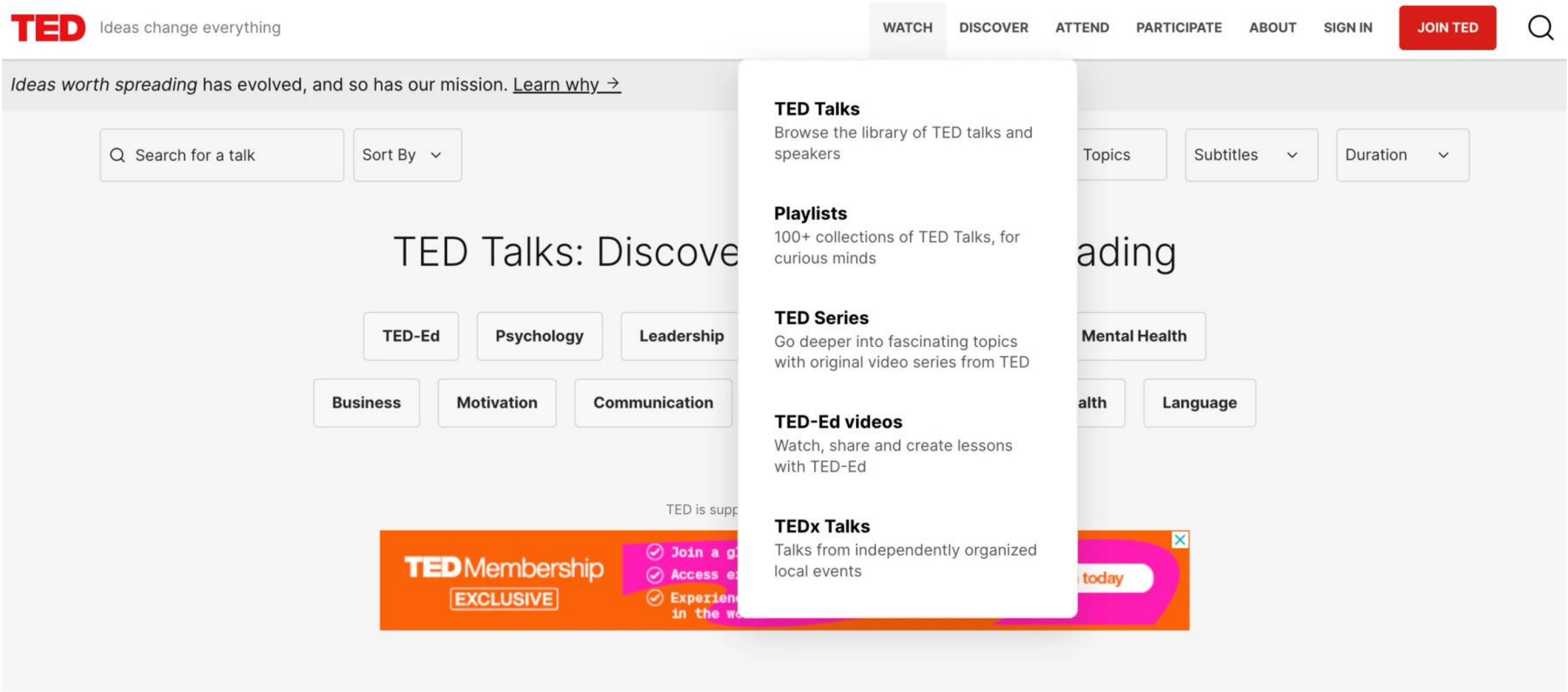
Pop-up: When pressed, command key for relevant options

Contextual: Provides access to often-used commands associated with a particular item. For example, right-clicking a photo on a website results in a small set of relevant menu options

Collapsible: Toggles between **+** and **–** icons on a header to expand or contract its contents

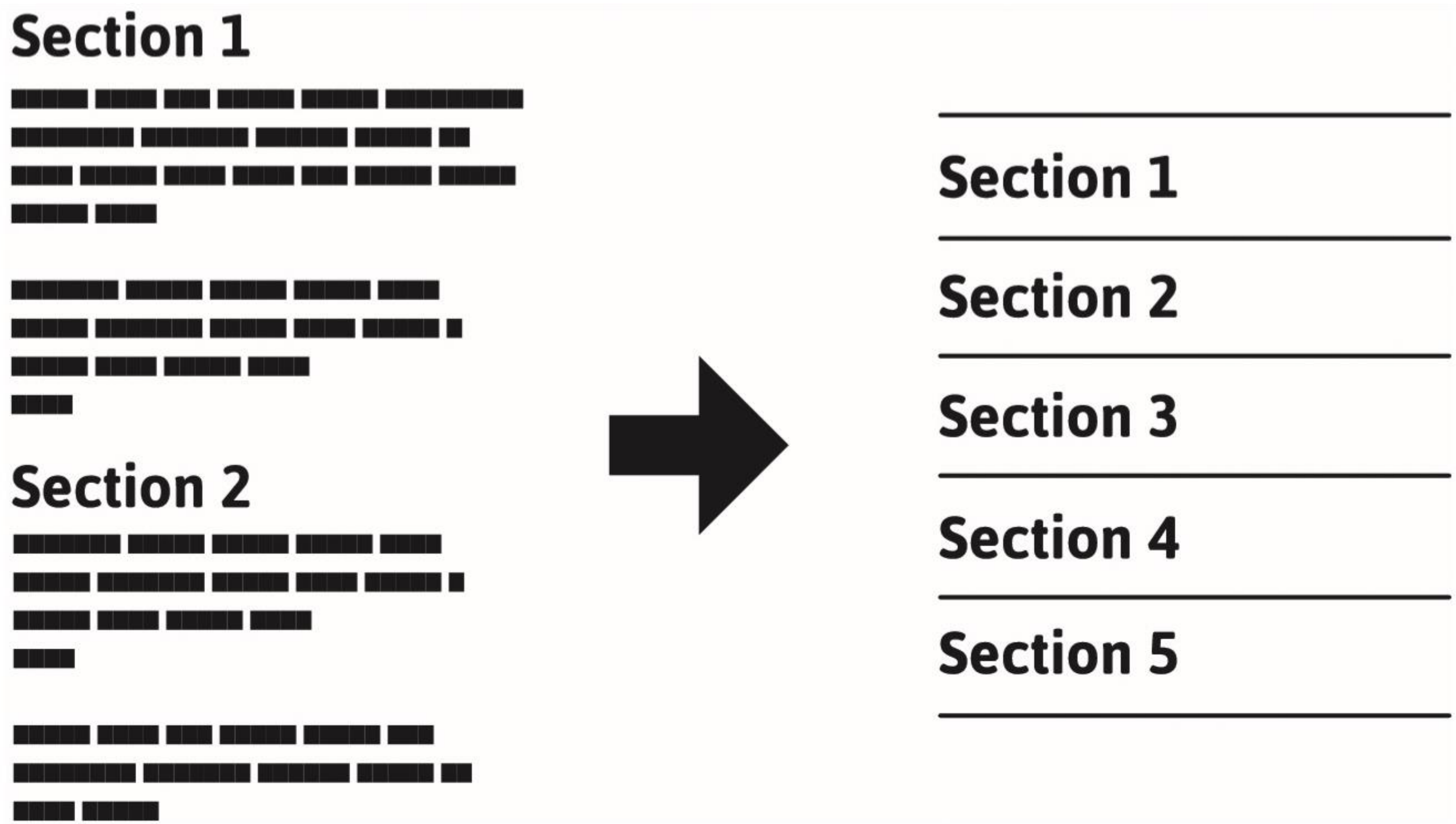
Mega: All options shown using 2D drop-down layout

A dropdown menu

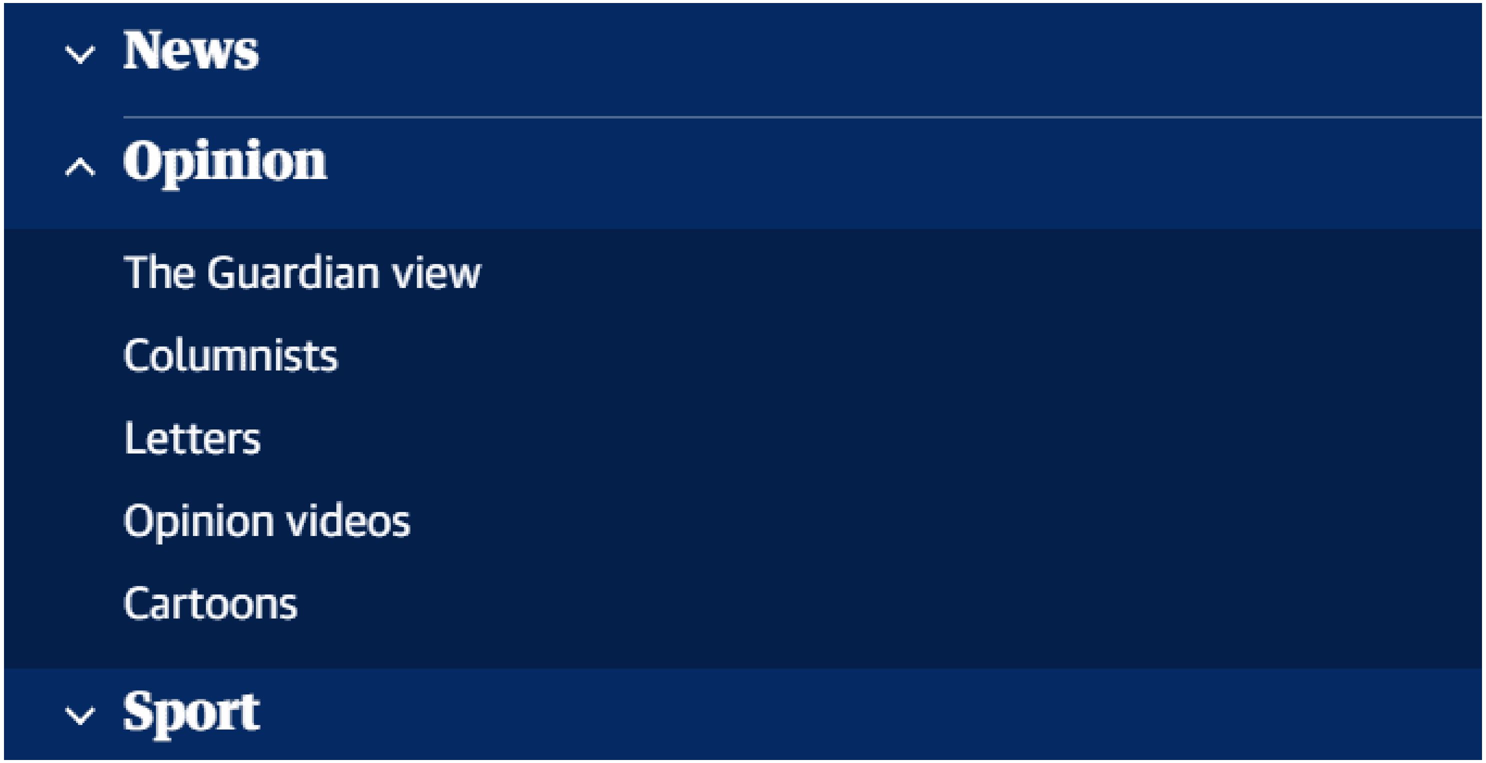


The TED website’s dropdown menu is narrow and has a single column listed vertically—typical of a regular dropdown menu.

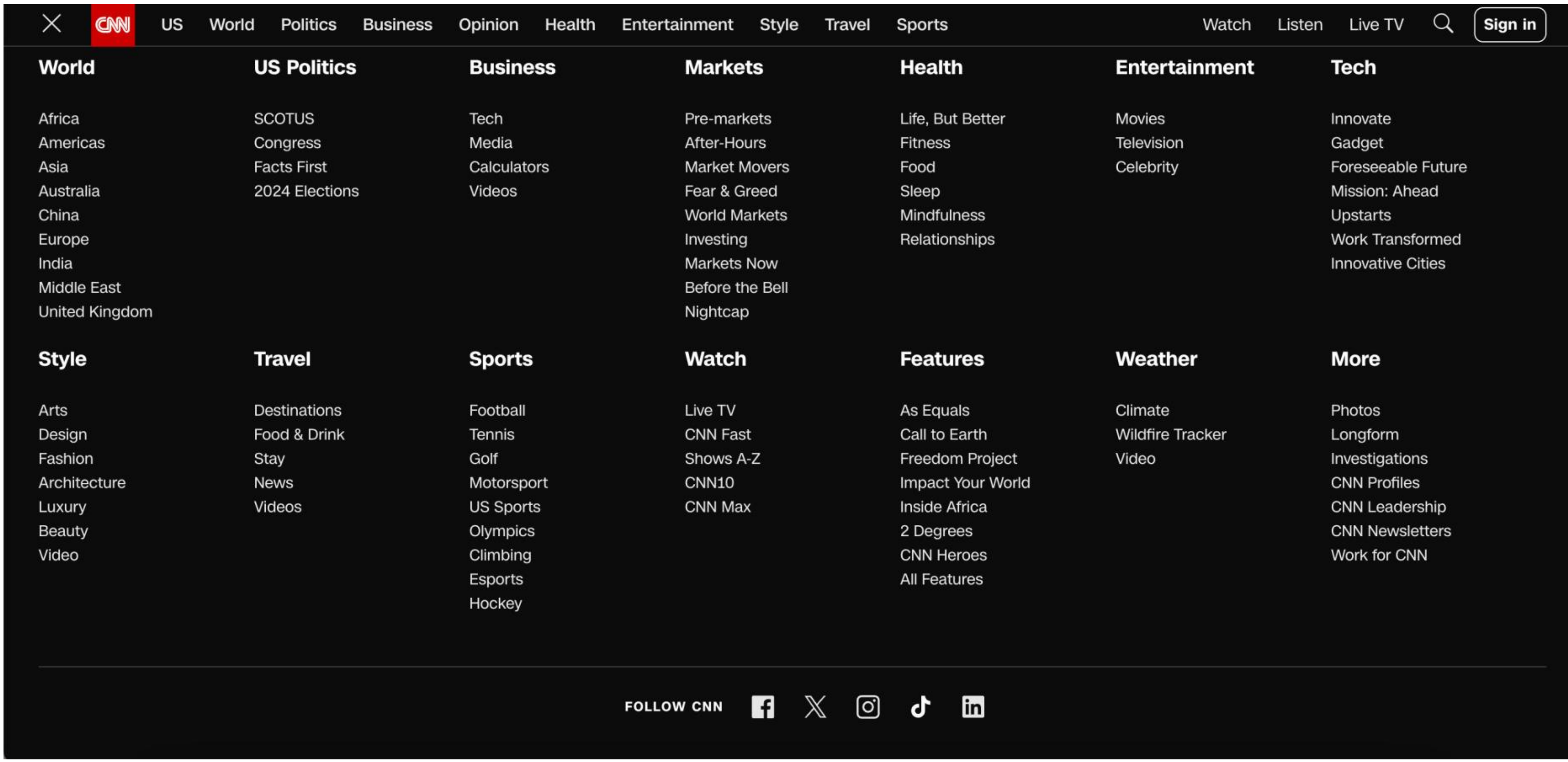
Template for a collapsible menu



The Guardian news outlet uses a collapsible menu for its mobile site

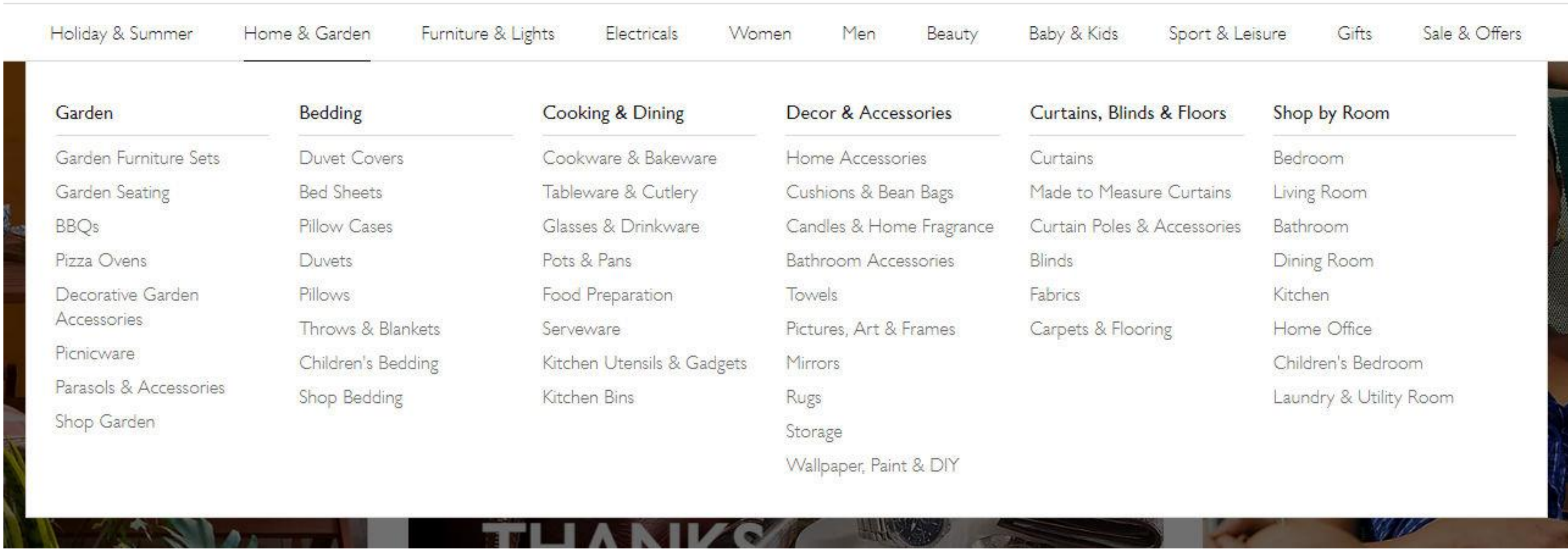


A Mega Menu



Websites with a lot of content, like CNN, opt for mega menus due to their complexity and extensive information architecture.

A Mega Menu



Mega menus can be intimidating, so meaningful grouping is important.

Design Considerations

- Design principles of spacing, grouping, and simplicity should be used
- Which terms to use for menu options (for example, “front” versus “bring to front”
- Mega menus easier to navigate than drop-down ones

Icon design

- Icons are assumed to be easier to learn and remember than commands
- Icons can be designed to be compact and variably positioned on a screen
- Now pervasive in every interface
  - For example, they represent desktop objects, tools (for example, a paintbrush), applications (for instance, a web browser), and (such as cut, paste, next, accept, and change) operations

Icons

- Since the Xerox Star days, icons have changed in their look and feel:
  - black and white →

    Color, shadowing, photorealistic images, 3D rendering, animation, flat

- Many designed to be very detailed and animated making them both visually attractive and informative

- Can be highly inviting, emotionally appealing, and feel alive

Icon forms

- The mapping between the **icon** and underlying **referent** (the object or operation to which the icon refers to) can be:
    - Similar (for example, a picture of a file to represent the object file)
    - Analogical (for instance, a picture of a pair of scissors to represent 'cut')
    - Arbitrary (such as the use of an X to represent 'delete')
- The most effective icons are those that are from the first and the 2nd categories
- Many operations are actions making it more difficult to represent them
    - Use a combination of objects and symbols that capture the salient/key part of an action

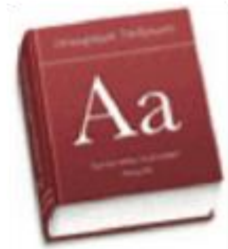
2 types of Apple Aqua icon styles

- Colorful photorealistic images were used in the original Apple Aqua set from 2000,
  - each slanting slightly to the left, for the category of user applications (such as email),
  - whereas monochrome straight on and simple images were used for the class of utility applications (for instance, printer setup).
  - The former has a fun feel to them, whereas the latter has a more serious look about them.

2 types of Apple Aqua icon styles

User Applications





Utility Applications




Two styles of Apple icons used to represent different kinds of functions

Flat 2D icons for a smartphone and a smartwatch

- Another approach that many smartphone designers use is flat 2D icons.
  - These are simple and use strong colors and pictograms or symbols. The effect is to make them easily recognizable and distinctive.
  - Examples shown on the following slide include
    - the white ghost on a yellow background (Snapchat),
    - a white line bubble with a solid white phone handset in a speech bubble on a lime-green background (WhatsApp), and
    - the sun next to a cloud (weather).

Flat 2D icons for a **smartphone** and a smartwatch

(a)

(b)

2D icons designed for
(a) a smartphone and
(b) a smartwatch

Flat 2D icons for a smartphone and a **smartwatch**

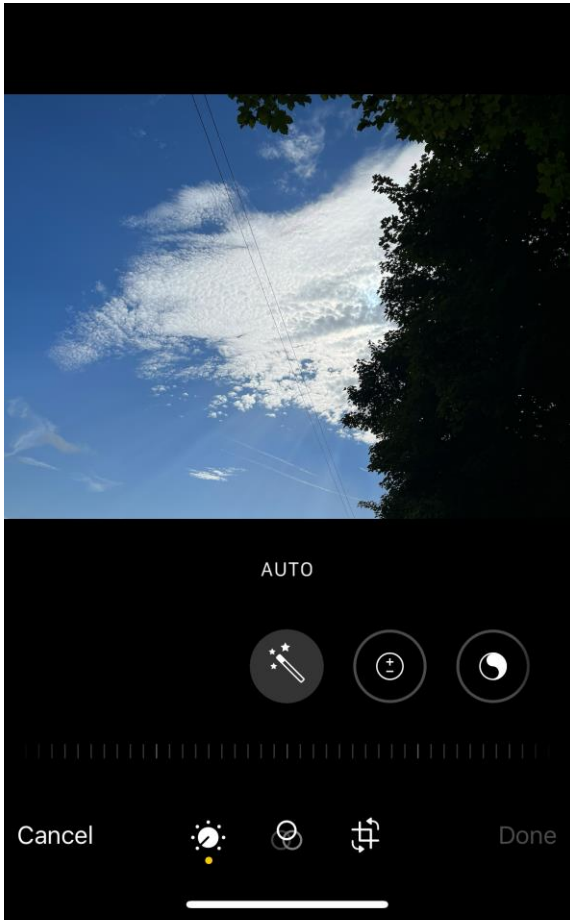
- **Icons** that are presented on small device displays (such as digital cameras or smartwatches) have much less screen space available.
- Because of this, they have been designed to be simple, emphasizing the outline form of an object or symbol and using only grayscale or one or two colors

Exercise

- Sketch simple icons to represent the following operations to appear on a digital camera screen:
  - Turn image 90-degrees sideways
  - Crop the image
  - Auto-enhance the image

- Show them to someone else and see if they can understand what each represents

Basic edit icons that appear on the iPhone app



The two rows of basic Edit Photo icons that appear at the bottom of an iPhone display

- The box with extended lines and two arrows is the icon for cropping an image
- the three overlapping translucent circles represents "different lenses" that can be used
- the wand above means "auto-enhance"
- the circle with plus and minus signs refers to exposure levels
- the circle to the right of it with the simplified ying and yang symbol refers to brilliance levels

Design considerations (3)

- There is a wealth of resources for creating icons
  - Guidelines, style guides, icon builders, libraries, online tutorials
- Text labels can be used alongside icons to help identification for small icon sets
- For large icon sets (for instance, photo editing or word processing) can use the hover function

Multimedia

- Combines different media within a single interface with various forms of interactivity
  - Graphics, text, video, sound, and animation
- Users click on a link in an image or text
  - An animation or a video clip is played
  - Users can return to where they were or move on to another place
- A combination of media and interactivity can provide better ways of presenting information than can a single media

Multimedia

- For example
  - KidsDiscover.com apps has many tablet apps that use a combination of animations, photos, interactive 3D models, and audio to teach kids about science and social studies topics.
  - Using swiping and touching, kids can reveal, scroll through, select audio narration, and watch video tours.
- Following figure, for example, has a “slide” mechanism as part of a tablet interface that enables the child to do a side-by-side comparison of what Roman ruins looks like now and in ancient Roman times.

Multimedia learning app designed for tablet

Pros and cons

- Pros:
  - Facilitates rapid access to multiple representations of information
  - Can provide better ways of presenting information than can any media alone
  - Can enable easier learning, better understanding, more engagement, and more pleasure
  - Can encourage users to explore different parts of a game or story
- Cons:
  - Tendency to play video clips and animations while skimming through accompanying text or diagrams

Design Considerations

- Multimedia good for supporting certain activities, such as browsing, but less optimal for reading at length

Virtual reality

- Computer-generated graphical simulations providing:
  - "the illusion of participation in a synthetic environment rather than external observation of such an environment" (Gigante, 1993)
- Provide new kinds of experience, enabling users to interact with objects and navigate in 3D space
- Create highly-engaging user experiences

https://www.interaction-design.org/literature/topics/virtual-reality

Pros and cons

- Can have a higher level of fidelity with objects that they represent compared to multimedia
- Induces a sense of presence where someone is totally absorbed by the experience
  - “a state of consciousness, the (psychological) sense of being in the virtual environment” (Slater and Wilbur, 1999)
- Provides different viewpoints:
  - First-person perspectives:
    - A person moves through the environment without seeing themselves. E.g., first-person shooter games such as DOOM
  - Third-person perspective:
    - Enables the person to see the virtual world above and behind their avatar

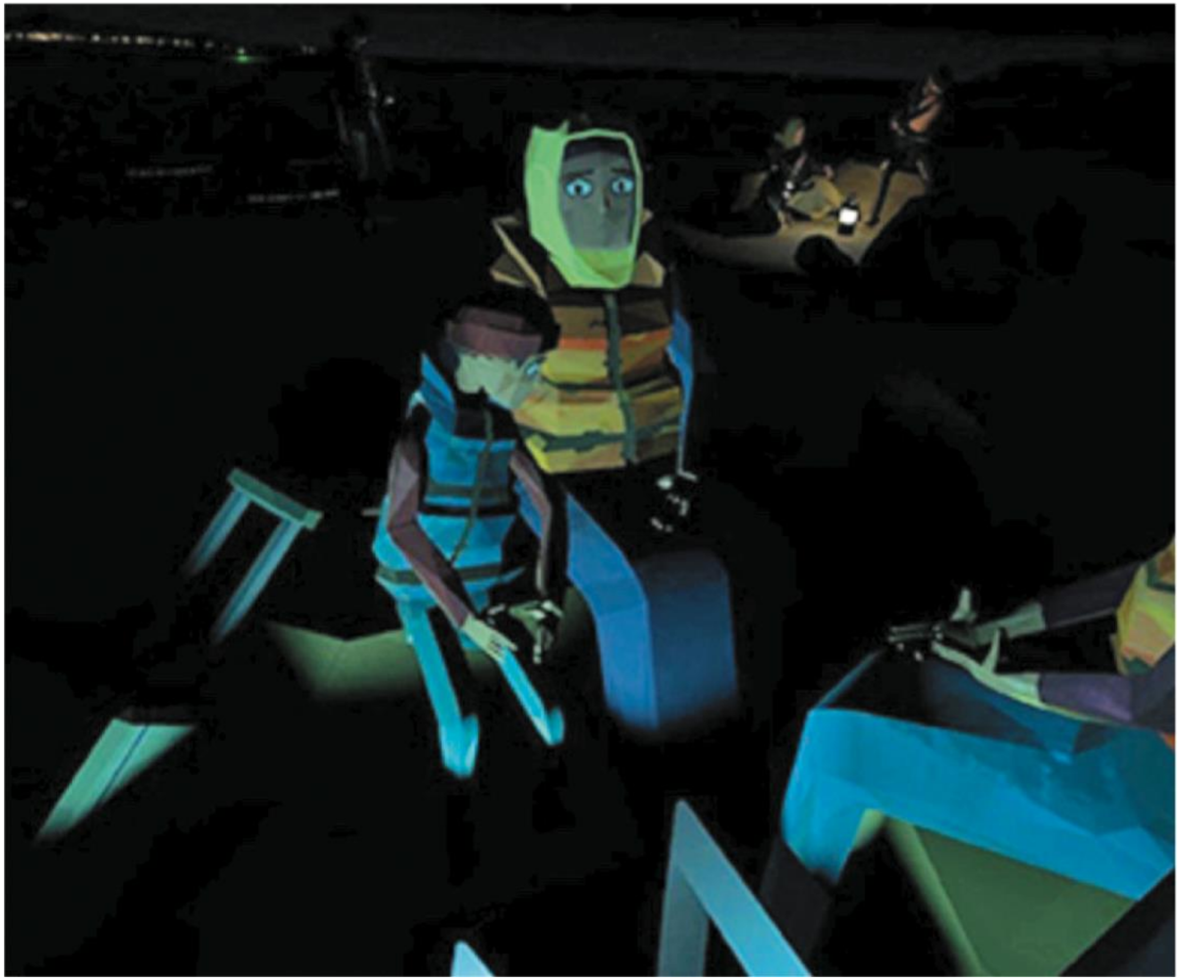
Pros and cons

- Early head-mounted displays were uncomfortable to wear and could cause motion sickness and disorientation
- Lighter VR headsets are now available (for example, HTC Vive) with more accurate head tracking

Polygon graphics used to represent avatars for the "We Wait" VR experience

The viewer is put on a boat with a group of refugees crossing the Mediterranean Sea

The goal was to let participants experience how it felt to
be there on the boat with the refugees.



Design considerations

- How best to navigate through VR environment (for instance, first versus third person)
- How to control interactions and movements (for example, by using head and body movements)
- How best to interact with information (for instance by using keypads, controllers, pointing, and joystick buttons)
- Level of realism to aim for to engender a sense of presence

Website design

- Early websites were largely text-based, providing hyperlinks
- Focus was on usability:
  - That is how best to structure information to enable users to navigate and access them easily and quickly
- Nowadays, more emphasis is on attractiveness
  - That is making pages distinctive, striking, and aesthetically pleasing
- Need to think of how to design information for different platforms—keyboard or touch?
  - For example, smartphones, tablets, and PCs

Usability versus aesthetics?

- Vanilla or multi-flavor design?
  - Ease of finding something versus aesthetic and enjoyable experience
- Web designers are:
  - "thinking great literature"
- Users read the web like a:
  - "billboard going by at 60 miles an hour" (Krug, 2014)
- Need to determine how to brand a web page to catch and keep attention

Breadcrumbs for navigation

Breadcrumbs are category/page labels:

- Enable users to look at other pages without losing track of where they have come from
- The breadcrumbs metaphor signifies the idea of leaving a trail or path to follow back.
- Enable one-click access to higher site levels
- Attract first time visitors to continue to browse a website having viewed the landing page

Breadcrumbs for navigation

- A breadcrumb trail on the Best Buy website showing three choices made by the user to get to Smart Lights

- Breadcrumbs consist of clickable links, each representing a higher-level category or page within the website's structure.

Web design styles

- how to design web browsers and websites for smaller-sized displays.
- A commonly used method is responsive website design
- **Responsive website design**
  - the browser automatically resizes the layout, and changes the graphic design, fonts, and appearance depending on the screen size (smartphone, tablet, or PC) on which it is being displayed.
  - This includes presenting the content over more pages
  - downsizing the content in this way makes it more time-consuming as more pages need to be loaded.
  - can also make it more fiddly navigating multiple pages and menus

Web design styles

- Another web design style is infinite scrolling
- **Infinite scrolling**
  - websites are designed to enable browsing content on one long page
    - Examples: social media sites, e-commerce, pinterest, tiktok, etc.
  - avoids a visitor needing to wait for pages to load when clicking on them
  - navigation is largely done by swiping across or down the page until the end is reached
  - a side effect is the tendency to glance while scrolling without focusing on individual items

Design considerations

- Many books and guidelines on website design
- Veen's (2001) three core questions to consider when designing any website:
  1. Where am I?
  2. Where can I go?
  3. What's here?

Exercise

- Look at a fashion brand's website,
  - for example, Nike.com or Levis
- What kind of website is it?
- How does it contravene/violate the design principles outlined by Veen?
- Does it matter?
- What kind of user experience is it providing for?
- What was your experience of engaging with it?

Mobile interfaces

- Handheld devices intended to be used while on the move
- Have become pervasive, increasingly used in all aspects of everyday and working life
  - For example, phones, fitness trackers, and smartwatches
- Customized mobile devices are also used by people in different work settings
  - For example, in restaurants to take orders, at car rental agencies to check in car returns, in supermarkets for checking stock.

Mobile interfaces

- Larger-sized tablets used in mobile settings
  - many airlines provide their flight attendants with one so that they can use their customized flight apps while airborne and at airports;
  - sales and marketing professionals also use them to demonstrate their products or to collect public opinions

iBeer App

- Designed for fun
- It detects the tilting of the iPhone and uses this information to mimic a glass of beer being consumed.
- The graphics and sounds are also very enticing;
- The beer can be drained if the phone is tilted enough

iBeer app

QR codes and smartphones

- QR (quick response) codes that store URLs and look like black-and-white checkered squares
  - provides quick access to relevant information
- A person focuses their camera phone over the QR code for a few seconds
- A notification appearing at the top or bottom of their screen
- When this is tapped, it takes them to the website

QR codes and smartphones

QR codes and smartphones

- This method became popular during the COVID-19 pandemic in cafes, bars, and restaurants
  - where customers selected their menu choices by first scanning the QR code on a card attached to their table and then scrolling through the available menu options on their phone.
- One of the main benefits of this approach was that it reduced human contact during service.
  - customers could also use it to order and pay for their meal.

Connecting to an online menu by scanning a QR code adhered to a table in a restaurant

Design considerations

- Mobile interfaces can be cumbersome to use for those with poor manual dexterity or ‘fat’ fingers
- Key concern is hit area:
  - Size of the area on the phone display that the user touches to make something happen, such as a key, an icon, a button, or an app
  - Space needs to be big enough for all fingers to press accurately
  - If too small, the user may accidentally press the wrong key
  - Fitts’ law can be used to help design right spacing
    - Minimum tappable areas should be 44 points x 44 points for all controls

Appliances

- Appliances include machines for everyday use in the home
  - for example, washing machines, microwave ovens, refrigerators, toasters, bread makers, and smoothie makers.
- And personal devices
  - For instance, digital clock and digital camera
- Used commonly for short periods of time to get something specific done
  - For example, starting the washing machine, watching a program, buying a ticket, changing the time, or taking a snapshot
- Need to be usable with minimal, if any, learning

Exercise

- Look at the controls on your toaster and describe what each does.

- Consider how these might be replaced with an LCD screen.
- What would be gained and lost from changing the interface in this way?

A typical toaster with basic physical controls

- Standard toasters have two main controls
  - a lever to press down to start the toasting and
  - a knob to set the amount of time for the toasting
- Many come with a small eject button that can be pressed if the toast starts to burn.
- Some also come with a range of settings for different ways of toasting (such as one side, frozen, and so forth), selected by moving a dial or pressing buttons.